

LYMPHOID NEOPLASIA

BCL11B enhancer hijacking by t(14;16)(q32;q24) translocation defines a novel high-risk subtype of T-ALL

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KEY POINTS

- We identified a novel T-ALL subtype with poor outcomes, overexpressing *FENDRR*, *FOXF1*, and *FOXC2* through *BCL11B* enhancer hijacking.
- This subtype is characterized by frequent *GATA3* mutations, marked lineage ambiguity, and enrichment in adolescents and young adults.

The molecular classification of T-cell acute lymphoblastic leukemia (T-ALL) remains incomplete, limiting risk stratification and the development of targeted therapies. Enhancer hijacking is a critical oncogenic mechanism that deregulates proto-oncogenes by repositioning cisregulatory regions via structural variants. Here, we performed an integrated analysis of pediatric and adult T-ALL and mixed-phenotype acute leukemias (MPALs), using whole-genome and whole-transcriptome sequencing. This analysis identified a group of 14 patients with predominantly T-lineage neoplasms driven by a t(14;16)(q32;q24) translocation, harboring universal *GATA3* mutations and *CDKN2A/B* deletions. Mechanistically, this translocation repositions the ThymoD locus downstream of *BCL11B*, causing monoallelic, ectopic overexpression of *FENDRR* and mesenchymal transcription factor genes *FOXF1* and *FOXC2* and activating epithelial-mesenchymal transition transcription signatures. Immunophenotypic and single-cell RNA sequencing analyses revealed marked lineage ambiguity with myeloid and B-cell differentiation potentials specific to this subtype. Furthermore, functional analyses in CD34⁺ cord blood cells demonstrated that *FOXF1* overexpression promotes myeloid differentiation while suppressing T-cell differentiation, serving as a key factor for lineage specification. Clinically, this subtype was detected in 0.15% to 4.0% of T-ALL/MPAL

cases depending on the cohort, showing a median age of 15 years and enrichment in adolescents and young adults. Importantly, patients with t(14;16)(q32;q24) have an extremely poor prognosis, showing a trend toward worse outcomes than high-risk groups such as KMT2A-rearranged early T-cell progenitor-like, SPI1-rearranged, and LMO2 $\gamma\delta$ -like T-ALLs. The unique molecular landscape and poor prognosis of patients with the t(14;16)(q32;q24) translocation underscore the need for the development of novel subtype-specific therapeutic approaches.

Introduction

T-cell acute lymphoblastic leukemia (T-ALL) is a hematological malignancy affecting both children and adults.^{1,2} Although our understanding of T-ALL has advanced significantly with the advent of next-generation sequencing of the genome and transcriptome,^{1,3-5} a consensus on its subclassification has yet to be reached. A refined understanding of T-ALL subtypes is crucial for developing stratified treatment approaches and personalized therapies.

One pivotal oncogenic mechanism of T-ALL is enhancer hijacking, in which structural variants (SVs) reposition active enhancers near proto-oncogenes, leading to their aberrant expression. A classic example involves chromosomal translocations that place the proto-oncogene *TAL1* under the control of the *SIL* enhancer, leading to its overexpression in developing T cells.⁶ Another example is the *BCL11B*-activated subtype, in which SVs activate *BCL11B* expression through tandem amplifications or juxtaposition with enhancers such as the *ARID1B*, *CCDC26*, and *ZEB2* regions.^{7,8} Conversely, the upregulation of *TLX3* via juxtaposition with the *BCL11B* enhancer is observed in *TLX3*-rearranged T-ALLs.^{9,10}

In this study, we identify and characterize a novel T-ALL subtype defined by the t(14;16)(q32;q24) translocation. This rearrangement results in *BCL11B* enhancer hijacking, which aberrantly activates the expression of the forkhead box (FOX) gene cluster¹¹ located in proximity to the breakpoint in 16q24. This cluster includes genes such as *FENDRR*, *FOXF1*, *FOXL1*, and *FOXC2*, which have not previously been implicated in hematological malignancies. We performed an integrated analysis of whole-genome sequencing (WGS), whole-transcriptome sequencing (WTS), and assay for transposase-accessible chromatin sequencing (ATAC-seq) to elucidate the molecular pathogenesis underlying this subtype.

Methods

Patients

A total of 14 patients with t(14;16)(q32;q24) were identified and included in this study. Four patients (P01-P03 and P11) were recruited specifically for this study. The screening of a previously published cohort by Seki et al³ (n = 121) identified 3 patients (P04-P06). An additional 3 patients (P12-P14) were identified by screening our in-house adult T-ALL/myeloid mixed-phenotype acute leukemia (MPAL) cohort (n = 74), which consisted of 66 T-ALL and 8 MPAL cases. The remaining patients (P07-P10) were identified in external data sets.

Samples were collected after obtaining written informed consent under protocols approved by the institutional review board of each institution. Patients P02, P03, and P12 were sequenced within the framework of the Genome Research in Cancers and Rare Diseases project as part of the AMED Program of Practical Research for Innovative Cancer Control (since fiscal year 2021; <https://www.amed.go.jp/en/program/list/15/01/010.html>).

Sorting of lymphoid and myeloid cell fractions of the bone marrow sample from patient P01

The bone marrow sample from patient P01 at diagnosis was stained with an allophycocyanin-conjugated anti-CD7 antibody

(catalog no. 653311; BD) and a phycoerythrin-conjugated anti-CD13 antibody (catalog no. 560098; BD). Dead cells were excluded via 7-amino-actinomycin D (catalog no. 559925; BD) gating. Lymphoid cells were defined as CD7⁺/CD13⁻, whereas myeloid cells were defined as CD7⁻/CD13⁺. In total, 4646 lymphoid cells and 24 911 myeloid cells were isolated and subjected to WGS.

Genomic sequencing

Genomic DNA libraries were prepared for WGS using the Illumina DNA PCR-Free Prep, Tagmentation (Illumina) and IDT for Illumina DNA/RNA UD Indexes (IDT). Subsequently, 150-bp paired-end sequencing was performed using the Illumina NovaSeq platform. For sorted lymphoid/myeloid samples of patient P01, genomic DNA samples were prepared for WGS using xGen cfDNA & FFPE DNA Library Preparation Kit (IDT) and 150-bp paired-end sequencing was performed using the MGI DNBSEQ sequencer. Whole-exome sequencing was prepared using the SureSelect XT Library Prep Kit and SureSelect XT Human All Exon V6 (Agilent), and 150-bp paired-end sequencing was performed using HiSeq2500 (Illumina). Raw reads were trimmed for adapter sequences using cutadapt (version 5.0)¹² and were subsequently aligned to the hg38 reference using bwa (version 0.7.17).¹³ Picard (GATK version 4.4.0)¹⁴ was used to mark duplicates.

Nanopore sequencing

Targeted adaptive sampling long-read sequencing was applied to genomic regions chr14:98223766-98796840 and chr16:86259172-86473017 (hg38), combined with a panel of 597 hematological malignancy-related genes. Sequencing was performed using a GridION or PromethION 2 Integrated sequencer and analyzed as previously described.^{15,16}

Patient whole-transcriptome sequencing (WTS)

RNA extraction and library preparation were performed using NEBNext Ultra RNA Library Prep kit or the TruSeq RNA Library Preparation Kit v2 and IDT for Illumina TruSeq RNA UD Indexes v2. Sequencing was performed using the Illumina NovaSeq platform. Reads were aligned to the hg38 reference genome and the GENCODE version 31 GTF using STAR (version 2.7.11a).¹⁷

ATAC-seq

The Fast-ATAC protocol was adopted as previously described.^{18,19} Reads were aligned to the hg19 reference genome using bowtie2 (version 2.3.3).²⁰ Subsequently, duplicates were removed by Picard (version 2.6.0). Peaks were called using HMMRATAC,²¹ followed by read count quantification using the multicov function in bedtools (version 2.27.1).²² edgeR²³ was used to identify peaks with differential accessibility, with a false discovery rate (FDR) threshold of 0.05. The enrichment of transcription factor-binding motifs was determined by the TOBIAS pipeline²⁴ using the JASPAR 2024 CORE nonredundant database.²⁵

FISH

SureDesign Dual Fusion FISH custom probes (Agilent) were designed to cover all the breakpoints on chromosomes 14 (fluorescein isothiocyanate-conjugated; chr14:97749139-99272346) and 16 (cyanine5-conjugated; chr16:85759376-87009247)

identified in this study (genomic coordinates in hg38). Hybridization and staining were performed on the formalin-fixed pericardial fluid specimen of P01 at diagnosis and a control normal lymph node, following the manufacturer's protocols. Images were acquired using BZ-X1000 (Keyence) with Plan Apochromat 100× Oil BZ-PA100 lens (Keyence).

Cell culture

Human cord blood (CB) cells were obtained from the Japanese Red Cross Kanto-Koshinetsu Cord Blood Bank. Informed consent was obtained, and all experiments were performed according to an institutional review board–approved protocol (approval number: 2025-45-0912), in accordance with the Declaration of Helsinki and The Belmont Report.

For stem cell culture, the cells were incubated in StemSpan SFEMII (Stemcell Technologies), supplemented with 50 ng/mL of recombinant human stem cell factor (SCF) (catalog no. 300-07; PeproTech), recombinant human thrombopoietin (catalog no. 300-18; PeproTech), and murine Flt3 ligand (Flt3L, catalog no. 250-31L; PeproTech) and 1% penicillin-streptomycin (catalog no. 09367-34; Nacalai). For myeloid differentiation culture, the cells were cultured in StemSpan SFEMII supplemented with 10 ng/mL of SCF, thrombopoietin, Flt3L, interleukin-3 (IL-3), and IL-6. For T-cell differentiation culture, coculture with OP9/N-DLL1 cells (RIKEN BRC Cell Bank: number RBRC-RCB2927)²⁶ was performed in minimum essential medium- α containing 20% fetal bovine serum supplemented with 50 ng/mL of rhSCF, 10 ng/mL of hIL-7 (catalog no. 200-07; PeproTech), and mFLT3L.

Plasmids

The *FOXF1* (NM_001451.3), *FOXC2* (NM_005251.3), and *FOXL1* (NM_005250.3) genes were cloned into the retroviral pMYs-MCS-IRES-EGFP vector.²⁷

External data sets

WGS data were obtained from either the database of Genotypes and Phenotypes (dbGaP) under accession number phs002276.v2.p1 and the Kids First data portal (<https://portal.kidsfirstdrc.org/dashboard>) or the St. Jude Cloud²⁸ (<https://www.stjude.cloud>). Gene expression data of the pediatric T-ALL cohort study by Pölönen et al⁴ was downloaded from Synapse (<https://doi.org/10.7303/syn54032669>). The single-cell RNA sequencing data set of the developmental thymus²⁹ was downloaded from the Human Developmental Cell Atlas (<https://developmental.cellatlas.io/studies/thymus-development>).

Additional methods

More detailed descriptions of the procedures are provided in the supplemental Methods (available on the *Blood* website).

Results

Identification of a novel t(14;16)(q32;q24) subtype in T-lineage leukemias

Our investigation was prompted by a 15-year-old male (index case P01) who presented with dyspnea and lower leg edema (Figure 1A). An echocardiogram revealed pericardial effusion, and a computed tomography scan showed a mediastinal mass. Flow cytometry of the pericardial effusion and bone marrow

identified a population of abnormal T-lineage cells with the following immunophenotype: negative for surface CD3, positive for cytoplasmic CD3, negative for myeloperoxidase, negative for CD1a, negative for CD8, negative for CD5, and positive for myeloid markers CD13 and CD11b. Initially, the patient was diagnosed with and treated for early T-cell precursor lymphoblastic lymphoma (ETP-LBL).³⁰ Despite 2 weeks of treatment with prednisolone, vincristine, daunorubicin, and L-asparaginase, leukocytosis persisted. Subsequent bone marrow examination revealed granulocytic hyperplasia with maturation, dysplastic changes in the myeloid lineage, and 10% lymphoblast-like cells (Figure 1B-C; supplemental Table 1). Consequently, the patient's diagnosis was revised to T/myeloid MPAL³⁰ with myeloproliferative neoplasm-like features in the myeloid lineage. Accordingly, the treatment was switched to an acute myeloid leukemia–type regimen. After 4 months of therapy, the mediastinal mass resolved, and the patient achieved a complete remission. However, 4 months after the completion of therapy, a combined duodenal and brain relapse occurred. Following a second complete remission, the patient underwent umbilical CB transplantation at month 17 and has remained in remission for over 6 years.

To determine the molecular basis of the index case, we performed WGS of bulk pericardial effusion cells and sorted lymphoid and myeloid cell fractions from the bone marrow specimen at diagnosis. All the samples shared a clonal origin with a t(14;16)(q32;q24) translocation (Figure 1D; supplemental Figure 1A-B) and a nearly identical set of somatic driver mutations (supplemental Table 2). The chromosome 14 breakpoint was located downstream of the *BCL11B* locus, a region associated with TLX3-translocated T-ALLs.^{1,9} The chromosome 16 translocation partner resided near the FOX gene cluster,¹¹ which had not been previously described in hematological malignancies, and prompted a systematic search for additional cases. Notably, no specific genomic difference was found to explain the lymphoid vs myeloid lineage commitment.

Through the analysis of available ALL samples and sequencing data, we identified a group of 14 patients with t(14;16)(q32;q24) translocations (supplemental Figure 1C-L; supplemental Table 3). This group, comprising 10 patients from our study and 4 from the literature,^{4,31} shared a distinct computational landscape characterized by universal GATA3 mutations and CDKN2A/B deletions (Figure 1E-F; supplemental Tables 4-7). For cases in which WGS was unavailable, targeted adaptive sampling via nanopore sequencing rapidly captured the t(14;16)(q32;q24) translocation.^{15,32} This method identified 5 patients P04, P06, P11, P13, and P14, 4 of whom (P04, P11, P13, and P14) were subsequently subjected to WGS.

The clinical presentation was predominantly T-ALL/LBL, diagnosed in 11 of the 14 patients. However, the group also included 2 T/myeloid MPAL cases (P01 and P12) and 1 B-LBL case (P03). Both MPAL cases were bilineage, showing distinct distributions between the T-cell (CD3⁺) and myeloid (myeloperoxidase-positive) tumor populations (supplemental Figure 2).

Given that both chromosome 14 and 16 breakpoints resided adjacent to the termini of the chromosomes, the t(14;16)(q32;q24) translocation was undetectable by conventional G-banding (supplemental Table 3). We therefore

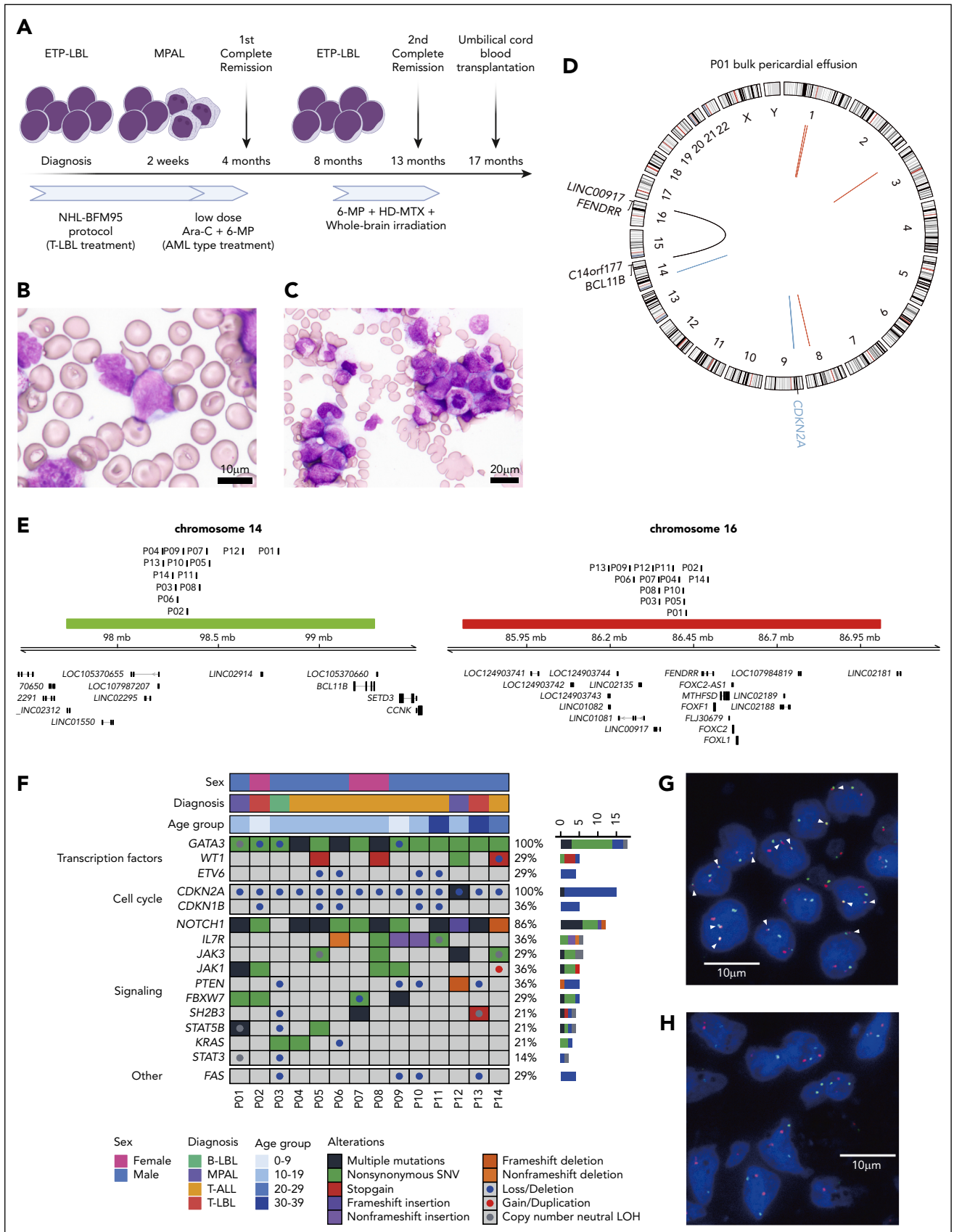


Figure 1. A novel recurrent T-ALL subtype defined by the t(14;16)(q32;q24) translocation. (A) Schematic diagram illustrating the clinical course of the index patient (P01). (B-C) Representative images of lymphoid (B) and myeloid (C) blasts identified by Giemsa staining of the bone marrow aspirate of patient P01 2 weeks after diagnosis. (D) Circos plot showing the SV profile of the pericardial effusion sample from patient P01 at disease diagnosis. The links represent deletions (blue), duplications (red), and translocations (black). For deletions and duplications, labels are shown only for genes included in the mutational landscape (panel F). For translocations, the genes nearest

designed FISH probes specific to t(14;16)(q32;q24), which successfully identified fusion signals in the index case P01 but not in a control patient (Figure 1G-H).

Transcriptome and ATAC-seq analysis of cases with t(14;16)(q32;q24) translocations

To evaluate the expression profiles of the t(14;16)(q32;q24)-driven T-ALLs, we performed WTS. Integrating our data with a published T-ALL data set⁴ (batch-corrected using ComBat-seq³³), uniform manifold approximation and projection analysis revealed that the t(14;16)(q32;q24) samples formed a distinct cluster positioned near ETP-like subtypes (Figure 2A; supplemental Table 8). Furthermore, ATAC-seq was performed on 3 patients (P01, P12, and P13) and 12 controls (supplemental Table 9) and supported the unique nature of this group, showing a distinct chromatin accessibility profile (Figure 2B; supplemental Figure 3).

This cluster was characterized by markedly elevated expression of genes within the FOX gene cluster¹¹ in proximity to the chromosome 16 breakpoint: *FENDRR*, *FOXF1*, *FOXC2*, and *FOXL1* (Figures 1E and 2C; supplemental Table 10). Allele-specific expression using heterozygous single-nucleotide polymorphisms in WTS data revealed monoallelic expression of *FOXF1* in all evaluable cases and of *FOXC2* in a subset of cases, whereas *FOXL1* did not show monoallelic expression (Figure 2D; supplemental Table 11). Importantly, the chromosome 16 breakpoint of patient P14 resided within the *FENDRR* gene body (Figure 1E), resulting in structural truncation and low expression level (supplemental Figure 4A-B; supplemental Table 10). Nevertheless, this patient showed the same disease phenotype, suggesting that *FENDRR* is dispensable for leukemogenesis. Finally, an in situ Hi-C library preparation followed by chromatin immunoprecipitation (HiChIP) assay of patient P01 confirmed a direct interaction between the *FOXF1*/*FENDRR* transcription start site and the ThymoD locus^{8,34} downstream of *BCL11B* (Figure 2E), indicating that this overexpression is driven by enhancer hijacking.

Concurrently, *BCL11B* itself was downregulated and exhibited monoallelic expression (Figure 2F-G; supplemental Tables 10 and 11), consistent with the loss of its cisregulatory enhancer on the translocated allele. Moreover, *BCL11B*-binding motifs were enriched in ATAC-seq peaks with decreased chromatin accessibility (Figure 2H), confirming the downregulation of the *BCL11B* transcriptional activity.

To elucidate the gene expression profiles of this newly defined subtype, we identified differentially expressed genes (in the t(14;16)(q32;q24) translocated T-ALLs compared with those in all other T-ALLs (supplemental Table 12). Subsequently, gene set enrichment analysis^{35,36} revealed that within the MSigDB hallmark gene sets,³⁷ the hallmark epithelial-mesenchymal transition (EMT) gene set was the most significantly enriched gene set in this subtype (Figure 2I; supplemental Table 13). Furthermore, to identify the key driver of the hallmark EMT

gene set enrichment, we performed overexpression of *FOXF1*, *FOXC2*, and *FOXL1* in CD34⁺ CB cells (supplemental Table 14). The EMT hallmark gene set was significantly enriched by *FOXF1* and *FOXC2* overexpression, whereas *FOXL1* overexpression did not show enrichment of the hallmark EMT gene set (Figure 2J-L). This is consistent with the established roles of *FOXF1* and *FOXC2* in EMT and mesenchymal tissue development.^{38,39}

In addition, within the ETP-like subtype, we performed gene set enrichment analysis after stratification based on the median expression of *FOXF1*, *FOXC2*, *FOXL1*, and *FENDRR* (supplemental Figure 5). The hallmark EMT gene set was significantly enriched in *FOXF1*- and *FOXC2*-high ETP-like T-ALLs, whereas *FENDRR*- and *FOXL1*-high T-ALLs did not show enrichment of the EMT gene set. These data corroborate the direct activation of EMT by *FOXF1* and *FOXC2* and further suggest that this mechanism is operative within the ETP-like subtype as well.

All these data suggest that the t(14;16)(q32;q24) translocation causes *BCL11B* enhancer hijacking, ectopically activating the 16q24 FOX gene cluster while downregulating *BCL11B* on the translocated allele. This new subtype is characterized by the activated EMT transcription program, likely by mesenchymal transcription factors *FOXF1* and *FOXC2*.

The distinct comutational landscape and intratumor heterogeneity of t(14;16)(q32;q24) T-ALL

Subsequently, we investigated the comutation profiles of this subtype. *GATA3* mutations were detected in all 14 cases. The most frequent alteration was in the R276 residue (p.R276L/Q), within the zinc finger (ZnF) domain (11/14 cases [78%]). Most of the remaining mutations were also located in the ZnF domain (Figure 3A). R276 mutations were significantly specific to this subtype compared with those in other *GATA3*-mutated T-ALLs in the study by Pölönen et al⁴ (5/23 cases [21%]; *P* = .0016, Fisher exact test). Furthermore, in the t(14;16)(q32;q24) subtype, *GATA3* mutations were frequently accompanied by either copy number loss, copy-number-neutral loss of heterozygosity, or multiple mutations occurring across alleles (50% vs 5% in other subtypes; *P* = .0021, Fisher exact test) (Figure 3B).

Somatic codriver mutations varied across T-ALL subtypes. Although t(14;16)(q32;q24) T-ALLs exhibited a high frequency of *GATA3* mutations, *PHF6* mutations were less common compared with those in *TLX3*-rearranged T-ALLs, and *FLT3* mutations/internal tandem duplications were absent, which are frequent in the *BCL11B*-activated subtype (Figure 3C). Although the *BCL11B* enhancer is involved in the pathogenesis of the latter 2 subtypes, the unique comutation profile reinforces the t(14;16)(q32;q24) subtype as a distinct entity.

To investigate intratumor heterogeneity and clonal evolution, we performed clonal evolution analysis on multiple samples from patients P01 and P14 (Figure 3D-E; supplemental

Figure 1 (continued) to the breakpoints are labeled. (E) Genomic distribution of translocation breakpoints in the 14 cases. The genomic regions covered by FISH probes used in panels G-H are shown with the corresponding green and red colors. (F) Mutational landscape of T-ALLs with the t(14;16)(q32;q24) translocation. (G) Representative FISH image of the pericardial fluid specimen of P01 at diagnosis. An overview of interphase nuclei from patient P01 showing fusion signals (arrows), indicative of the translocation. (H) Representative FISH image of a lymph node of a fusion-negative control patient. Ara-C, cytarabine; HD-MTX, high-dose methotrexate; LOH, loss of heterozygosity; 6-MP, 6-mercaptopurine; SNV, single nucleotide variant. Figure 1A created with BioRender.com. Mimura K. (2025) <https://biorender.com/hcvj936>.

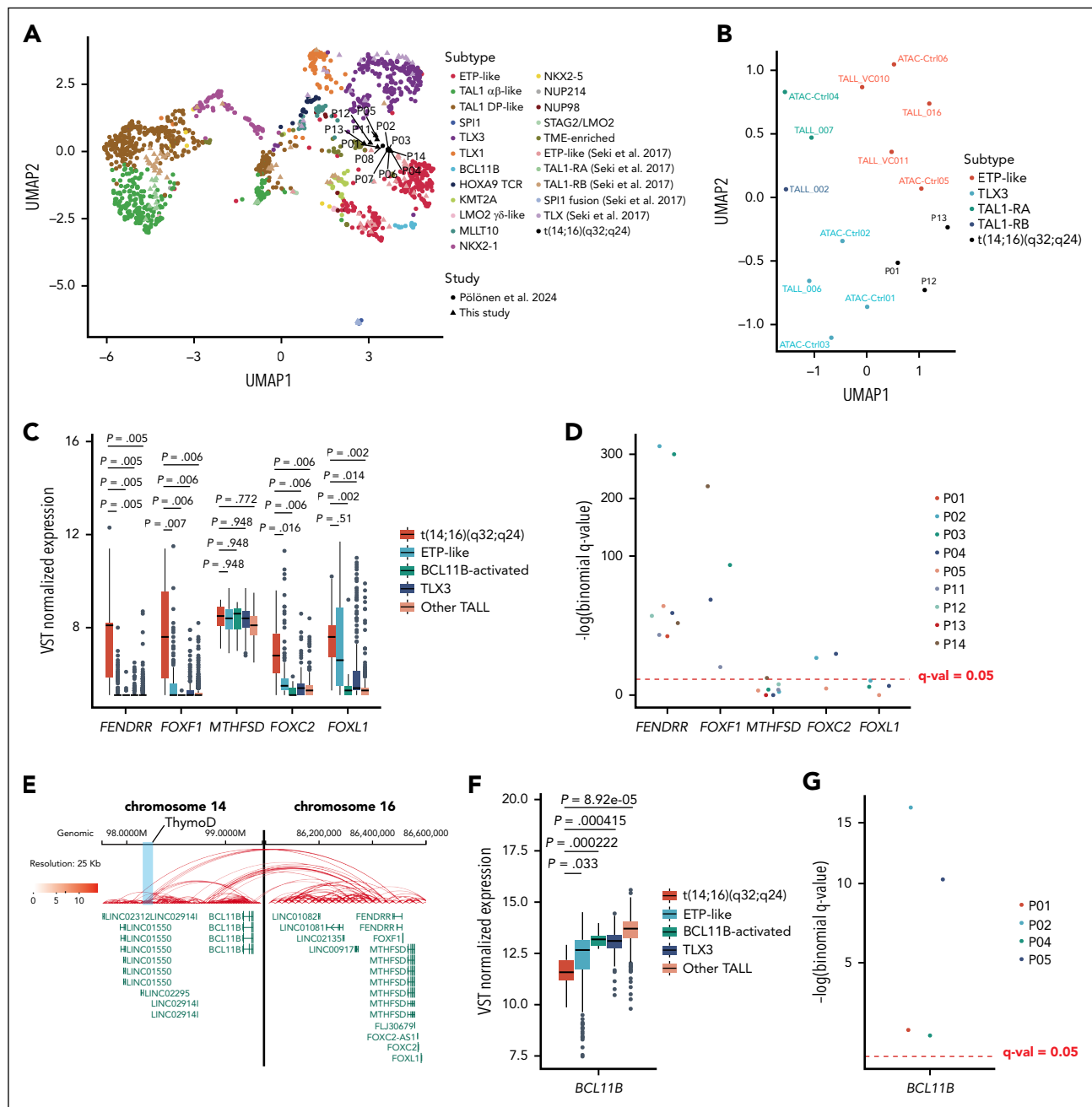


Figure 2. Gene expression profiles of T-ALL with the t(14;16)(q32;q24) translocation. (A) Uniform manifold approximation and projection (UMAP) projection of WTS data from a combined cohort of T-ALL samples. The data set includes samples from Pölonen et al.⁴ and this study, which were integrated and batch-corrected. T-ALL subtypes are labeled according to the classification by Pölonen et al.⁴ with specific sample annotations from Seki et al.³ where relevant. (B) UMAP projection of ATAC-seq data of T-ALL samples. (C) Box plot showing the variance-stabilizing transformation (VST)-normalized expression of genes located near the 16q24 breakpoint. The *P* value was calculated with the Welch *t* test in each comparison, followed by FDR adjustment. (D) Monoallelic expression of phased alleles of genes near the 16q24 breakpoint. Negative log-transformed *q*-values from a binomial test are plotted for each gene in each sample. The *y*-axis is on a square root scale. The red dotted line represents a *q*-value of 0.05. (E) HiChIP interaction track of patient P01. The ThymoD locus downstream of *BCL11B* is highlighted. (F) VST-normalized *BCL11B* expression across T-ALL subtypes. The *P* value was calculated with the Welch *t* test in each comparison, followed by FDR adjustment. (G) Monoallelic expression of *BCL11B*. Negative log-transformed *q*-values from a binomial test are plotted for each gene in each sample. The *y*-axis is on a square root scale. The red dotted line represents a *q*-value of 0.05. (H) The result of the TOBIAS pipeline with decreased chromatin accessibility in t(14;16)(q32;q24) T-ALLs. (I) Gene set enrichment analysis (GSEA) results showing the top MSigDB Hallmark gene sets enriched in the t(14;16)(q32;q24) T-ALL subtype. (J-L) GSEA enrichment plot for the HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION gene set, analyzing differentially expressed genes in CD34⁺ CB cells overexpressing *FOXF1* (J), *FOXC2* (K), and *FOXL1* (L). *q*-val, *q*-value.

Figure 6A-B; supplemental Tables 15 and 16). Patient P01 acquired 4 distinct *NOTCH1* mutations at 3 time points, whereas P14 acquired *NOTCH1* mutations on 2 separate occasions. Several other mutations in *FBXW7*, *WT1*, and the

JAK/STAT pathway were identified as subclonal events, frequently occurring as multiple mutations within a single gene, whereas *GATA3* mutations were uniformly clonal (supplemental Figure 6A-B). The contrast between the truncal

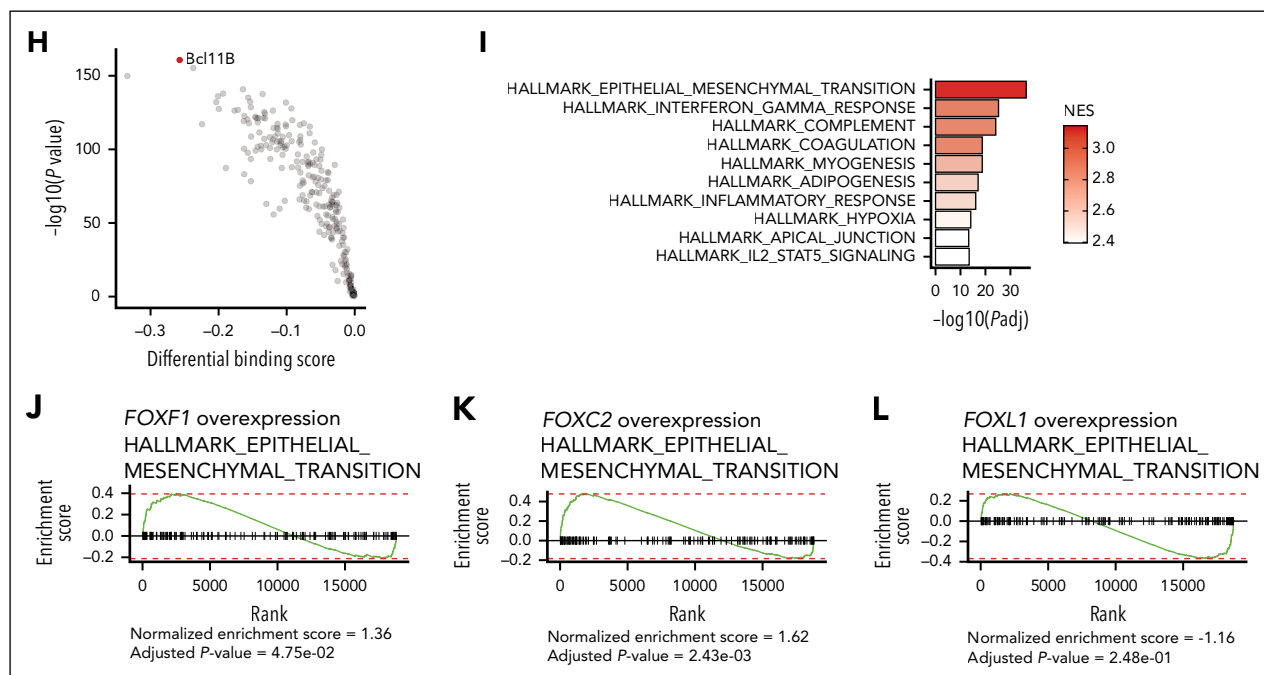


Figure 2 (continued)

GATA3 mutation and other subclonal driver mutations was consistent in other single-sampled patients as well (supplemental Figure 6C; supplemental Table 17).

Collectively, these findings underscore the specificity and truncal nature of GATA3 mutations in the t(14;16)(q32;q24) subtype, particularly those affecting the R276 residue or the ZnF domain. Following this, the tumor exhibited dynamic clonal evolution, acquiring late-stage mutations in NOTCH1, FBXW7, and the JAK/STAT pathway, often accumulating multiple mutations within the same gene.

The immature developmental origin and lineage ambiguity of t(14;16)(q32;q24) T-ALLs

To investigate the developmental origin of T-ALL with t(14;16)(q32;q24), we first analyzed the T-cell receptor (TCR) gene rearrangements (Figure 4A; supplemental Table 18). The absence of TCR rearrangements in all analyzed cases suggests that this subtype originates from an early hematopoietic progenitor before T-cell lineage commitment, an observation similar to previous reports on the BCL11B-activated subtype.⁸

However, our subsequent analyses suggested a complex cellular identity characterized by profound lineage ambiguity rather than a simple early progenitor stage. Gene expression deconvolution using a single-cell RNA sequencing data set of the thymus²⁹ showed that t(14;16)(q32;q24)-rearranged T-ALLs possessed lower hematopoietic stem cell signatures than the BCL11B-activated subtype, suggesting a more differentiated state (Figure 4B; supplemental Table 19). This is consistent with low CD34 positivity in this subtype (Figure 4C; supplemental Table 20).

Most prominently, this subtype displayed a strong myeloid propensity. Gene expression deconvolution analysis revealed a

specific enrichment of signatures for monocytes, macrophages, and dendritic cells in the t(14;16)(q32;q24)-rearranged T-ALLs (Figure 4B). This finding was corroborated by immunophenotypic analysis, which showed high positivity of the myeloid marker CD13 (Figure 4C; supplemental Table 20). The identification of patients presenting with T/myeloid MPAL (P01 and P12) clinically validates this myeloid potential. Mechanistically, ATAC-seq confirmed increased chromatin accessibility at binding sites for key myeloid transcription factors, such as the CEBP family genes and SPI1 (PU.1) (Figure 4D).

In addition to myeloid features, this subtype showed evidence of B-cell differentiation potential. CD19 positivity was specific to this group (3/6 cases with available CD19 data). This may be explained by the significantly higher expression of SPI1 and its downstream target EBF1 (Figure 4E; supplemental Table 10), which form a critical axis for B-cell differentiation.⁴⁰ In addition, increased chromatin accessibility at EBF1-binding sites was observed (Figure 4D). Patient P03, who presented with B-LBL, may serve as a clinical example of this B-cell potential.

Subsequently, we evaluated the effect of gene overexpression on differentiation in CD34⁺ CB cells (Figure 4F). In OP9-DL1 coculture conditions, which normally support T-cell differentiation, FOXF1-overexpressing cells exhibited increased CD13 positivity, whereas CD7 positivity was suppressed compared with that in empty vector controls (Figure 4G). This myeloid skewing was consistently observed in FOXF1-overexpressing cells cultured in stem cell and myeloid differentiation culture conditions (supplemental Figure 7). In contrast, this myeloid bias was not observed in FOXC2-overexpressing cells in either culture condition (supplemental Figure 7). Collectively, the results identify FOXF1 as a key factor promoting myeloid

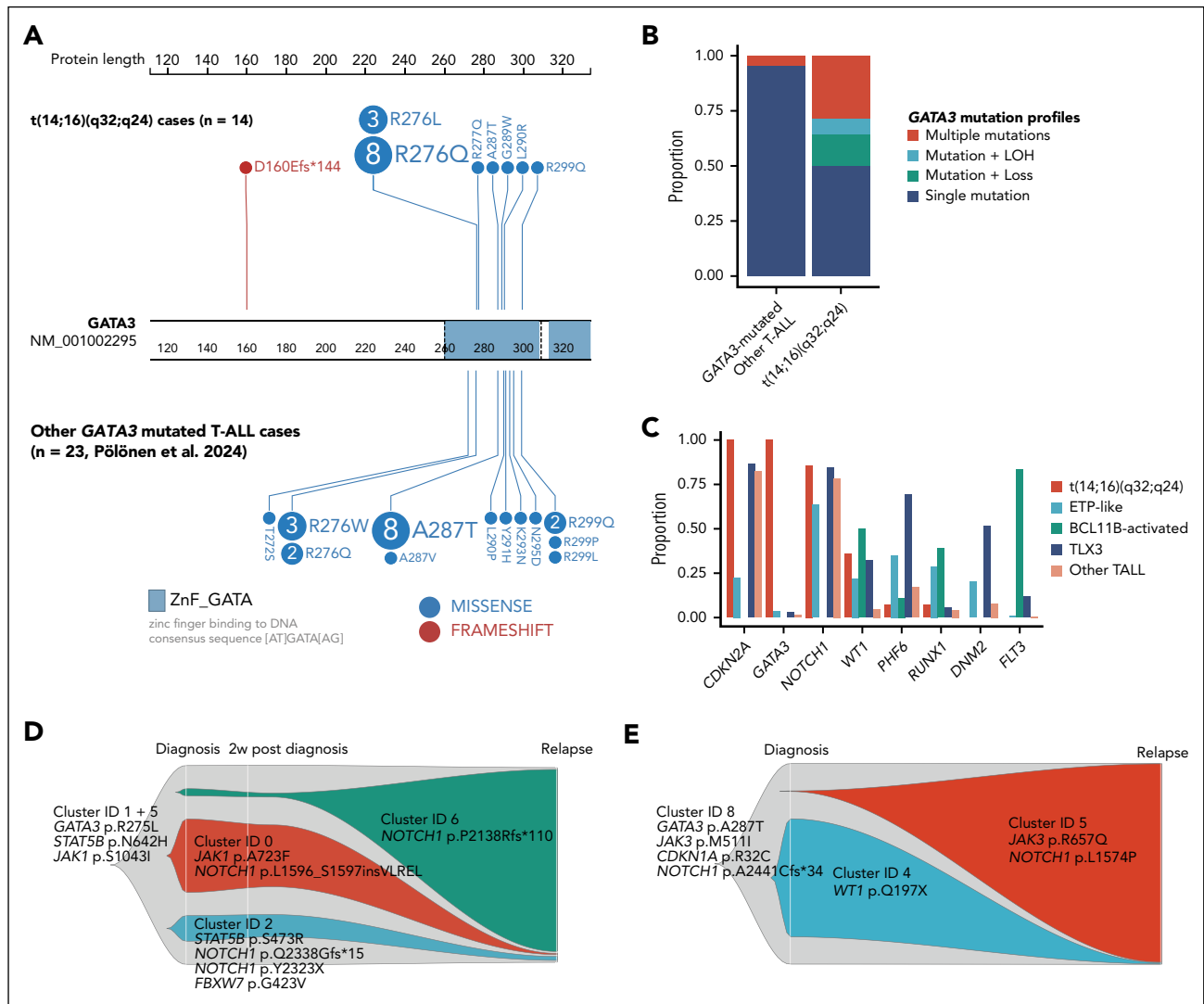


Figure 3. Distinct comutation landscape of T-ALL with the t(14;16)(q32;q24) translocation. (A) Distribution and types of GATA3 mutations detected in T-ALLs with the t(14;16)(q32;q24) translocation and other GATA3-mutated T-ALLs. (B) Bar chart comparing the proportion of cases with multiple GATA3 mutations between t(14;16)(q32;q24) T-ALLs and other GATA3-mutated T-ALLs. (C) Bar chart showing the frequency of driver mutations across subtypes. (D) Fish plot illustrating the clonal evolution in patient P01. Cancer cell fraction (CCF) was estimated at 3 time points: diagnosis (onset), 2 weeks post diagnosis, and relapse. (E) Fish plot illustrating the clonal evolution in patient P14. CCF was estimated at 2 time points: diagnosis and relapse. ID, identifier; LOH, loss of heterozygosity.

differentiation and suppressing T-cell differentiation in this subtype.

Collectively, the absence of TCR rearrangements, combined with the multilineage gene expression, immunophenotypic, and epigenetic profiles, indicates that T-ALL with t(14;16)(q32;q24) arises from an early hematopoietic progenitor with remarkable lineage ambiguity, exhibiting concurrent myeloid and B-cell characteristics. Notably, the myeloid bias observed in FOXF1-overexpressing CD34⁺ CB cells highlights FOXF1 as a key regulator of this lineage specification.

Poor prognosis associated with T-ALL defined by t(14;16)(q32;q24) translocation

We evaluated the demographic and clinical features of patients with this novel t(14;16)(q32;q24) entity. This subtype presented at an older age at diagnosis (median age: 15 years; Figure 5A; supplemental Table 3), compared with that in other T-ALL

subtypes in Pölönen et al (participants aged 1-30 years).^{4,41} The t(14;16)(q32;q24) group was predominantly male (11/14 [78%]; Figure 5B). The incidence varied across cohorts, identified in 2 of 1309 (0.15%) patients in the pediatric T-ALL cohort by Pölönen et al,⁴ 3 of 121 (2.4%) in the Japanese pediatric T-ALL cohort by Seki et al,³ and 3 of 74 (4.0%) in our in-house adult T-ALL/MPAL cohort. This reflects an enriched prevalence in the adolescent and young adult population.

Subsequently, we compared the survival outcomes with those reported in the previous study by Pölönen et al,⁴ which stratified T-ALL risk into 4 risk groups: "very high risk" (ETP-like^{KMT2A}, SPI1, and LMO2 $\gamma\delta$ -like subtypes), "high risk" (ETP-like^{MLLT10}, NUP98, Rare, TAL1 $\alpha\beta$ -like^{Other}, and NKX2-5 subtypes), "low risk," and "very low risk." Patients with the t(14;16)(q32;q24) translocation exhibited significantly poorer overall survival and event-free survival than the "high risk" T-ALL group (Figure 5C-D). Although not statistically different

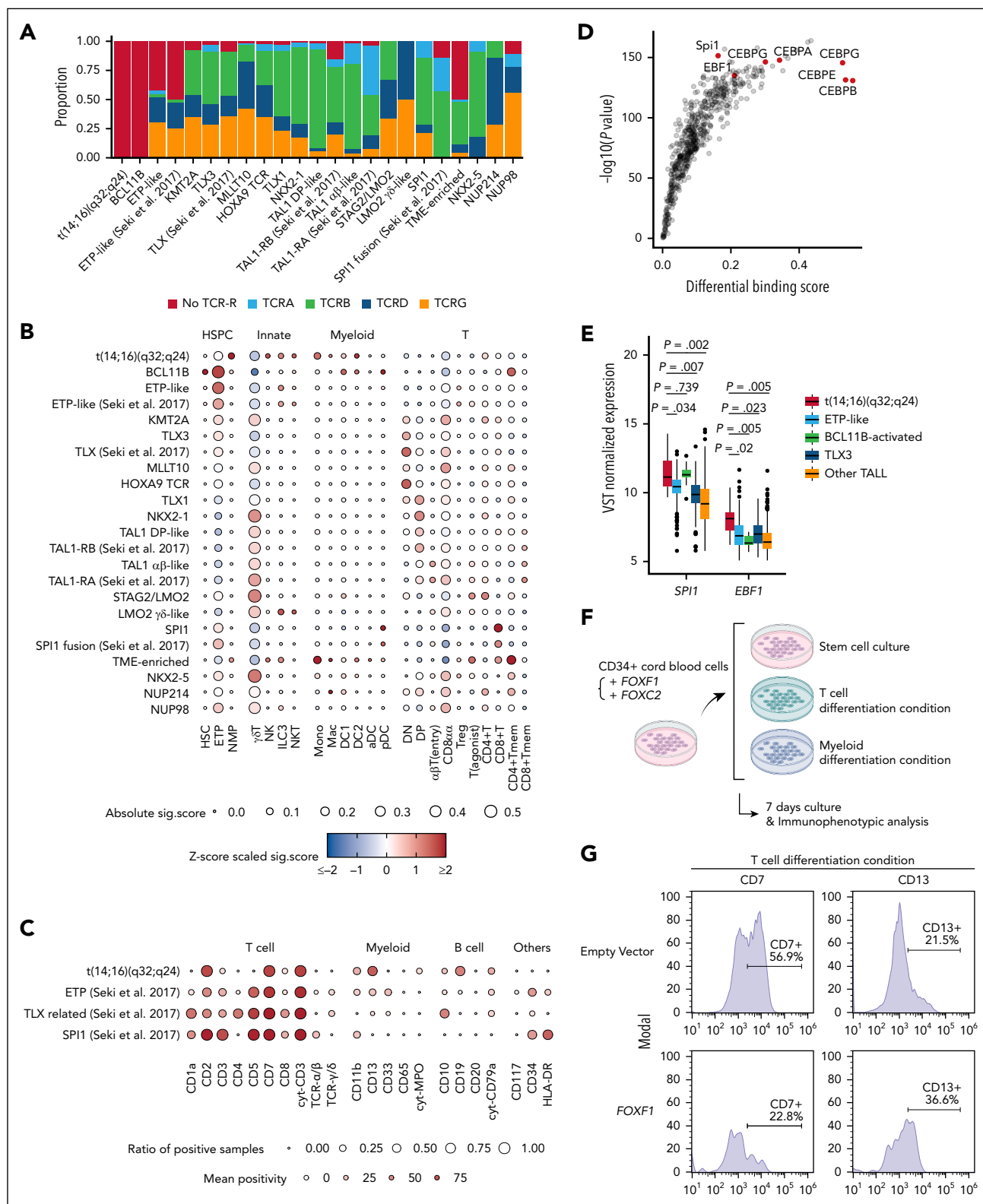


Figure 4. Lineage ambiguity of T-ALL with the t(14;16)(q32;q24) translocation. (A) Status of the TCR gene rearrangements across T-ALL subtypes. (B) Cell-type deconvolution of bulk WTS data. The plot shows the signature scores of thymic cell populations within each T-ALL subtype, inferred from a single-cell RNA sequencing data set of the thymus. The color represents the mean z score-scaled significance scores of each patient within the subtype, and the dot size represents the mean absolute significance scores of each patient within the subtype. Individual panel titles represent HSPC, innate T/lymphoid cells (Innate), myeloid cells (Myeloid), and T cells (T). (C) Dot plot summary of the flow cytometry (FCM) analysis. Within each T-ALL subtype, the dot size represents the percentage of samples positive for a given marker (defined as >50% positive cells). The color intensity corresponds to the mean percentage of positive cells within the samples. (D) The results of the TOBIAS pipeline on increased chromatin accessibility in t(14;16)(q32;q24) T-ALLs. (E) VST-normalized *SPI1* and *EBF1* expression across T-ALL subtypes. The *P* value was calculated with the Welch *t* test in each comparison, followed by FDR adjustment. (F) Schematic diagram of the immunophenotypic experiments performed using CD34⁺ CB cells. (G) FCM analysis result of

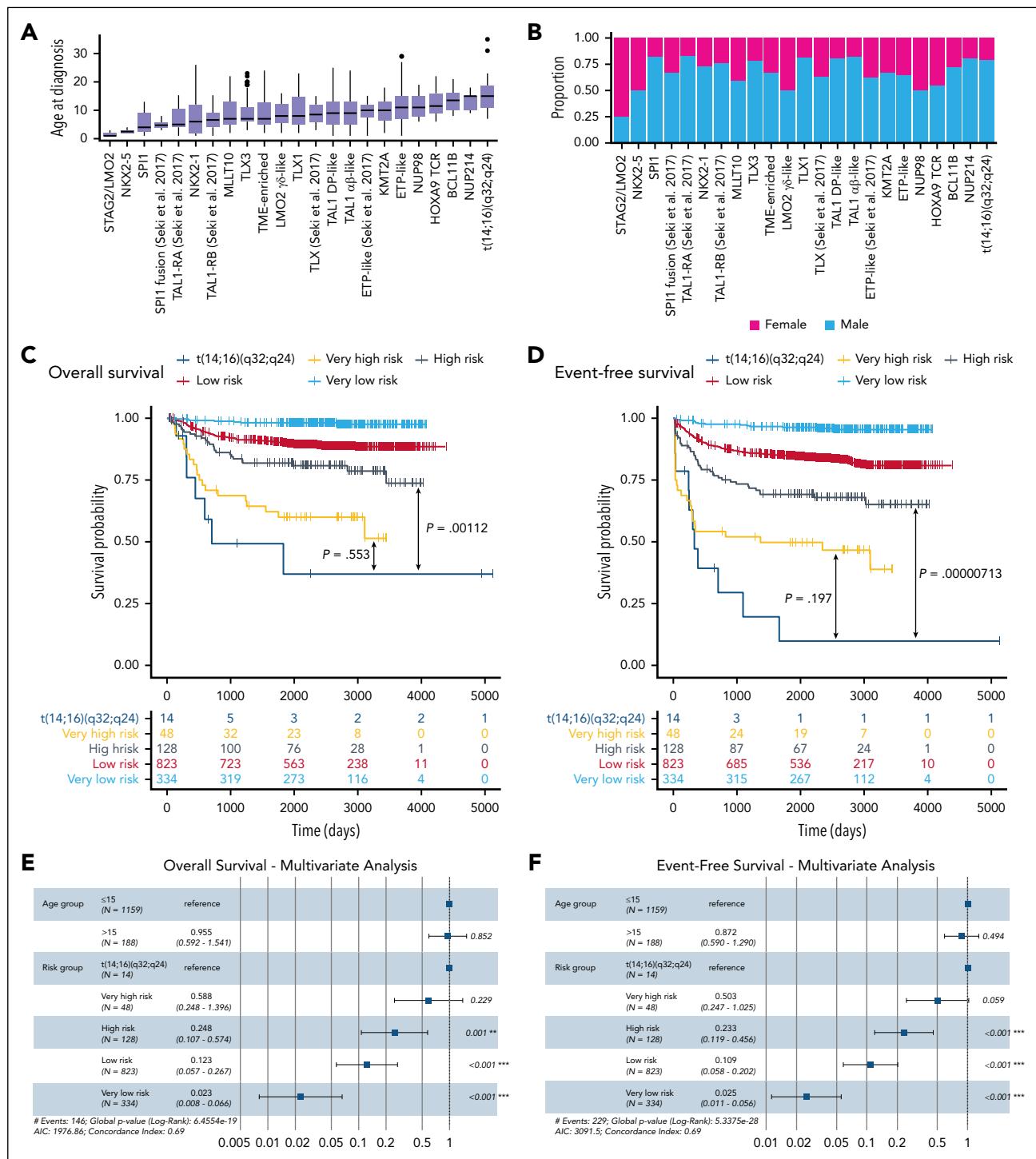


Figure 5. Poor prognosis and clinical characteristics of T-ALL with the t(14;16)(q32;q24) translocation. (A) Box plot comparing the age at diagnosis across T-ALL subtypes. (B) Bar chart showing the sex distribution within each T-ALL subtype. (C-D) Kaplan-Meier analysis comparing overall survival (OS) (C) and event-free survival (EFS) (D) between the t(14;16)(q32;q24) group and the Pölönen et al cohort, stratified by risk groups defined by Pölönen et al.⁴ Pairwise log-rank tests were performed between T-ALLs with t(14;16)(q32;q24) translocation and the “very high risk” and “high risk” T-ALL groups, respectively, followed by *P* value adjustment by the Bonferroni method. (E-F) Forest plot of the multivariate Cox proportional hazards analysis for OS (E) and EFS (F) between the t(14;16)(q32;q24) group and the Pölönen et al cohort. Both models include age and risk group as covariates. Hazard ratios and 95% confidence intervals are shown. AIC, Akaike Information Criterion.

Figure 4 (continued) CD34⁺ CB cells transduced with empty vector or FOXF1, cultured in T-cell differentiation condition for 7 days and stained using the markers shown. aDC, activated dendritic cells; cyt, cytoplasmic; DC, dendritic cell; DN, double negative; DP, double positive; HSPC, hematopoietic stem and progenitor cells; ILC3, innate lymphoid cell 3; Mac, macrophage; Mono, monocyte; NK, natural killer cell; NKT, natural killer T cell; NMP, neutrophil-myeloid progenitor; pDC, plasmacytoid dendritic cell; sig., significance; Tmem, memory T cell; Treg, regulatory T cell. Figure 4F created with BioRender.com. Mimura K. (2026) <https://biorender.com/ajw7bnw>.

from the “very high risk” group, the t(14;16)(q32;q24) group trended toward inferior survival, suggesting an aggressive disease course (Figure 5C-D).

Considering that adult ALL generally has poorer outcomes than pediatric ALL,² we performed subgroup analyses stratified by age (over/under 15 years) to exclude age distribution as a confounding factor. The prognostic impact remained consistent (supplemental Figure 8). Furthermore, multivariate analysis including age as a covariate confirmed the t(14;16)(q32;q24) translocation as an independent predictor of poor overall survival and event-free survival (Figure 5E-F). These results confirm the aggressiveness of this subtype irrespective of age, highlighting the need for intensive clinical management.

Discussion

Here, we report a novel T-ALL subtype characterized by poor clinical outcomes and a distinct computational landscape. By integrating WGS, WTS, and ATAC-seq, we defined its molecular features, demonstrating that the overexpression of mesenchymal transcription factors FOXF1 and FOXC2, not previously implicated in leukemia, drives an EMT-like phenotype. Furthermore, this T-ALL subtype exhibited lineage ambiguity, exhibiting concurrent myeloid and B-cell characteristics. Through overexpression experiments in CD34⁺ CB cells, we showed that FOXF1 promoted myeloid differentiation and suppressed T-cell differentiation.

The t(14;16)(q32;q24) group exhibited extremely poor outcomes: only 5 of 14 patients survived, all of whom had undergone allogeneic stem cell transplantation (supplemental Table 3). This suggests that allogeneic stem cell transplantation may be essential for this subtype under current treatment paradigms. For diagnosis, clinical sequencing can detect the distinct driver mutation profiles characterized by frequent GATA3 mutations and the absence of *PHF6* and *FLT3* mutations/internal tandem duplications. In this study, we utilized adaptive sampling via nanopore sequencing for screening,¹⁵ enabling rapid, bait-free detection of specific mutations and SVs. In addition, WTS can identify FOX gene cluster overexpression and distinct expression signatures, which may facilitate the diagnosis. Furthermore, we have demonstrated that FISH is another viable diagnostic option. Conversely, conventional G-banding likely fails to detect this translocation due to the breakpoint location adjacent to the termini of the chromosomes.

Although the role of EMT is well established in solid tumors, its significance in leukemias remains poorly understood. Recent evidence has shown that the EMT program drives drug resistance⁴² and invasiveness⁴³ in acute leukemias. In this subtype, we observed transcriptional changes indicative of EMT program activation, mediated by mesenchymal transcription factors FOXF1 and FOXC2. Furthermore, several co-occurring mutations found in this subtype may cooperate with the EMT program, suggesting novel functions for these mutations beyond their previously described roles in leukemia. For instance, *NOTCH1* mutations, known as potent EMT drivers in solid cancers,⁴⁴ were frequently observed. In addition, *WT1* mutations may also promote EMT.⁴⁵

The unique molecular landscape of this t(14;16)(q32;q24) subtype suggests several potential avenues for developing novel subtype-based therapeutic approaches. Clinically actionable alterations in this disease include *NOTCH1* mutations,^{46,47} IL7R/JAK/STAT pathway mutations,⁴⁸ and *CDKN2A* deletions.⁴⁹ In addition, the significant upregulation of hallmark mTORC1 signaling and phosphatidylinositol 3-kinase/Akt/mTOR signaling gene signatures (supplemental Table 13) suggest the potential utility of mTOR inhibitors. These agents have been shown to act synergistically with JAK inhibitors⁵⁰ and CDK4/6 inhibitors.⁵¹ The development of a novel, molecular-based therapeutic approach is critical to tackle this aggressive subtype.

A primary limitation of this study is its small sample size; larger, prospective studies are required to accurately determine the prevalence and fully characterize the clinical features of this disease. Survival comparisons are further limited by the treatment heterogeneity in our group (supplemental Table 3). In contrast, the previous study by Pölönen et al⁴ included patients treated according to the standardized protocol of the AALL0434 trial.⁵² This discrepancy in treatment regimens may confound direct prognostic comparisons. Furthermore, additional study is needed to fully understand the roles of mesenchymal transcription factors FOXF1 and FOXC2 in hematological malignancies, their impact on drug resistance, and the transcriptional activity of the R276-mutated GATA3 protein.

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Authorship

Contribution: K. Mimura and K.Y. designed the study; A. Kaino, M. Seki, Y.N., E.H., K. Mitani, H.K., J. Takita, Y.M., M. Shimomura, R.K., R.O., D.H., S. Katayama, H.N., Y. Sasahara, A. Kikuchi, A.S.-O., S. Kato, K.W., M.K., S.Y., N.H., K. Koh, and S.S. provided the DNA samples and/or clinical data; A.O., Y. Okamura, F.K., and K. Kinoshita performed exome sequencing for the P01 sample; K. Mimura, Y. Ochi, and M.H. performed the bioinformatics analysis; Y. Ochi and S.O. performed the assay for transposase-accessible chromatin sequencing and HiChIP assays; Y.-H.C., K. Mimura, and S.G. performed the functional assays; M.T. performed the fluorescence in situ hybridization assays; M.T.-O., H.M., and N.K. performed the sequencing experiments; A.S.-O., S. Kato, K.W., and M.K. performed nanopore sequencing; K. Katayama, S.I., and Y. Shiraishi processed the whole-genome sequencing data; K. Mimura, A. Kaino, J. Takeda, and K.Y. prepared the manuscript; and all authors participated in discussions and interpretation of the data and results.

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Footnotes

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Sequencing data generated within the Genome Research in Cancers and Rare Diseases (G-CARD) project have been deposited in the G-CARD Metadata Viewer (<http://vanilla.hgc.jp>; G-CARD study identifier [ID]: G-CARDS000005).

Other sequencing data generated in this study have been deposited in the European Genome-phenome Archive (<https://ega-archive.org/>; study ID: EGAS50000001254).

The remaining data are available from the corresponding author, Kenichi Yoshida (keyoshi2@ncc.go.jp), on reasonable request.

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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